Notes on Wavenumber-Domain Channel Estimation for UPA-Based HMIMO

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***Abstract*—This note summarizes a wavenumber-domain training and estimation framework for narrowband holographic MIMO (HMIMO) systems with uniform planar arrays (UPAs). [1] By leveraging Fourier plane-wave series (FPWS) dictionaries induced by finite apertures, we derive a compact hybrid input–output model and propose a selection-based probing strategy that enables sparse recovery in the wavenumber domain.**

***Index Terms*—Holographic MIMO, extremely large aperture arrays, Fourier plane-wave series, wavenumber domain, hybrid beamforming, channel estimation.**

I. Introduction

II. Temporal Wavenumber-Domain HMIMO Model and Problem Formulation

We consider a narrowband uplink or downlink HMIMO link operating at carrier frequency f_c, with wavelength $\lambda = c/f_c$ and wavenumber $k_0 = 2\pi/\lambda$. The transmitter and receiver are equipped with rectangular uniform planar arrays (UPAs) parallel to the xOy-plane. For $I \in \{R, S\}$, let $N_{I,x}$ and $N_{I,y}$ denote the numbers of antenna elements along the x- and y-axes, respectively, and define $N_r = N_{R,x}N_{R,y}$ and $N_t = N_{S,x}N_{S,y}$. The corresponding physical aperture lengths are

$$L_{I,x} = (N_{I,x} - 1)\delta, \qquad L_{I,y} = (N_{I,y} - 1)\delta, \tag{1}$$

where δ is the common inter-element spacing. Throughout this paper, we assume $\delta \le \lambda/2$ so that the propagating wavenumber spectrum is alias-free over the discretized aperture support.

A. Static FPWS-Based Channel Representation

Let $\boldsymbol{\xi}_R$ and $\boldsymbol{\xi}_S$ denote the receive- and transmit-side propagating mode sets induced by the Fourier plane-wave series (FPWS) expansion, and let $L_R = |\boldsymbol{\xi}_R|$ and $L_S = |\boldsymbol{\xi}_S|$. The spatial-domain channel matrix at time index t is represented as

$$\mathbf{H}_t = \boldsymbol{\Psi}_R \mathbf{H}_{a,t} \boldsymbol{\Psi}_S^H, \tag{2}$$

where $\boldsymbol{\Psi}_R \in \mathbb{C}^{N_r \times L_R}$ and $\boldsymbol{\Psi}_S \in \mathbb{C}^{N_t \times L_S}$ are the FPWS dictionaries, and $\mathbf{H}_{a,t} \in \mathbb{C}^{L_R \times L_S}$ is the discrete wavenumber-domain channel matrix.

To capture clustered scattering while preserving a physically meaningful second-order structure, we adopt the separable variance model

$$\mathbf{H}_{a,t} = \operatorname{diag}\big(\sqrt{\boldsymbol{\sigma}_{R,t}}\big)\, \mathbf{W}_t \operatorname{diag}\big(\sqrt{\boldsymbol{\sigma}_{S,t}}\big), \tag{3}$$

where $\boldsymbol{\sigma}_{R,t} \in \mathbb{R}_+^{L_R}$ and $\boldsymbol{\sigma}_{S,t} \in \mathbb{R}_+^{L_S}$ denote the receive- and transmit-side modal power profiles, respectively, and the entries of $\mathbf{W}_t \in \mathbb{C}^{L_R \times L_S}$ are zero-mean unit-variance complex Gaussian random variables. In this form, the dominant support of $\mathbf{H}_{a,t}$ is governed mainly by the slowly varying modal power profiles, whereas $\mathbf{W}_t$ captures the fast small-scale fluctuations.

B. Hybrid Measurement Model Over Time

At time index t, the channel is probed over P training slots. Let $\mathbf{Q}_{t,p} \in \mathbb{C}^{N_{\mathrm{RF}}^{(r)} \times L_R}$ and $\mathbf{x}_{t,p} \in \mathbb{C}^{L_S}$ denote the effective receive-side probing matrix and transmit-side probing vector at slot p, respectively, after projection onto the FPWS dictionaries. Then the slot-wise measurement is

$$\mathbf{y}_{t,p} = \sqrt{P_{\mathrm{dim}}}\, \mathbf{Q}_{t,p} \mathbf{H}_{a,t} \mathbf{x}_{t,p} + \mathbf{n}_{t,p}, \qquad p = 1, \ldots, P, \tag{4}$$

where $\mathbf{n}_{t,p} \sim \mathcal{CN}(\mathbf{0}, \sigma_t^2 \mathbf{I})$ and P_{dim} is the pilot-power normalization factor.

By stacking all slot-wise observations, we obtain

$$\mathbf{y}_t = \sqrt{P_{\mathrm{dim}}}\, \mathcal{A}_t(\mathbf{H}_{a,t}) + \mathbf{n}_t, \tag{5}$$

where $\mathbf{y}_t \in \mathbb{C}^M$ with $M = PN_{\mathrm{RF}}^{(r)}$, $\mathbf{n}_t \sim \mathcal{CN}(\mathbf{0}, \sigma_t^2 \mathbf{I}_M)$, and $\mathcal{A}_t(\cdot)$ denotes the linear sensing operator defined by

$$\mathcal{A}_t(\mathbf{Z}) \triangleq \begin{bmatrix} \mathbf{Q}_{t,1}\mathbf{Z}\mathbf{x}_{t,1} \\ \mathbf{Q}_{t,2}\mathbf{Z}\mathbf{x}_{t,2} \\ \vdots \\ \mathbf{Q}_{t,P}\mathbf{Z}\mathbf{x}_{t,P} \end{bmatrix}. \tag{6}$$

Letting $\mathbf{h}_t = \operatorname{vec}(\mathbf{H}_{a,t})$, (5) can be equivalently written as

$$\mathbf{y}_t = \mathbf{C}_t \mathbf{h}_t + \mathbf{n}_t, \tag{7}$$

where $\mathbf{C}_t$ denotes the implicit sensing matrix associated with $\mathcal{A}_t(\cdot)$.

C. Temporal Evolution and Groupwise State Variables

To model channel evolution in dynamic HMIMO environments, we assume that the large-scale modal structure varies slowly over time, whereas the small-scale term evolves with temporal correlation. Specifically, we adopt

$$\mathbf{W}_{t+1} = \rho_w \mathbf{W}_t + \sqrt{1 - \rho_w^2}\, \widetilde{\mathbf{W}}_{t+1}, \tag{8}$$

where $0 \le \rho_w < 1$ and $\widetilde{\mathbf{W}}_{t+1}$ has i.i.d. $\mathcal{CN}(0,1)$ entries. Meanwhile, the modal power profiles $\boldsymbol{\sigma}_{R,t}$ and $\boldsymbol{\sigma}_{S,t}$ evolve according to slowly drifting scattering clusters, so that the dominant support pattern of $\mathbf{H}_{a,t}$ is temporally correlated.

To explicitly exploit this structure, we partition the index set of $\mathbf{h}_t$ into G disjoint groups,

$$\{1, \ldots, L_R L_S\} = \bigcup_{g=1}^{G} \mathcal{I}_g, \qquad \mathcal{I}_g \cap \mathcal{I}_{g'} = \varnothing,\ g \ne g', \tag{9}$$

where each group corresponds to a localized block in the $(\ell_x, \ell_y, m_x, m_y)$ wavenumber grid. The energy of group g at time t is defined as

$$E_{g,t} \triangleq \left\|\mathbf{h}_{g,t}\right\|_2^2, \qquad \mathbf{h}_{g,t} \triangleq \mathbf{h}_t|_{\mathcal{I}_g}. \tag{10}$$

We further define the binary support indicator

$$b_{g,t} \triangleq \mathbf{1}(E_{g,t} \geq \tau_t)\,, \tag{11}$$

and the log-energy state

$$e_{g,t} \triangleq \log(E_{g,t} + \varepsilon), \tag{12}$$

where $\tau_t > 0$ is an energy threshold and $\varepsilon > 0$ is a small constant. The pair $\{b_{g,t}, e_{g,t}\}_{g=1}^{G}$ will serve as the low-dimensional predictive state of the time-varying channel.

D. Problem Formulation

In practical operation, the BS does not observe $\mathbf{H}_{a,t}$ directly. Instead, only the noisy hybrid measurements $\{\mathbf{y}_t\}$ and the probing configuration $\{\mathcal{A}_t\}$ are available. Our goal is to exploit the temporal correlation of the wavenumber-domain group structure to improve the recovery of the current channel. Specifically, given the past L frames

$$\mathcal{H}_t \triangleq \left\{\mathbf{y}_{t-L+1:t},\, \mathcal{A}_{t-L+1:t},\, \widetilde{\mathbf{H}}_{a,t-L+1:t}\right\}, \tag{13}$$

where $\widetilde{\mathbf{H}}_{a,\tau}$ denotes a coarse estimate at time τ, we aim to predict the next-step group support and group energy,

$$\left(\widehat{\mathbf{b}}_{t+1}, \widehat{\mathbf{e}}_{t+1}\right) = f_\Theta(\mathcal{H}_t), \tag{14}$$

and then use the prediction to warm-start the structured Bayesian recovery of $\mathbf{H}_{a,t+1}$ from $\mathbf{y}_{t+1}$. Therefore, the overall task is not to replace model-based estimation by a black-box predictor, but to couple temporal data-driven forecasting with physics-consistent operator-form channel recovery.

III. Wavenumber-Token Design and LLM-Based Predictive Support Learning

This section develops a lightweight causal sequence model that operates on physics-aware wavenumber tokens rather than raw high-dimensional CSI. The design objective is to predict the slowly varying group support and group energy in (11)–(12), which are then fed to the model-based recovery stage.

A. Coarse State Extraction

At each time index t, we first form a coarse channel estimate $\widetilde{\mathbf{H}}_{a,t}$ from the current measurements. The coarse estimate may be obtained, for example, by a matched-filter snapshot,

$$\widetilde{\mathbf{H}}_{a,t}^{(0)} = \sqrt{P_{\mathrm{dim}}}\,\mathcal{A}_t^*(\mathbf{y}_t), \tag{15}$$

or by running only a small number of iterations of a low-complexity recovery algorithm. From $\widetilde{\mathbf{H}}_{a,t}$, we compute the corresponding coarse group energies

$$\widetilde{E}_{g,t} = \left\|\widetilde{\mathbf{h}}_{g,t}\right\|_2^2, \qquad \widetilde{\mathbf{h}}_{g,t} \triangleq \mathrm{vec}(\widetilde{\mathbf{H}}_{a,t})|_{\mathcal{I}_g}, \tag{16}$$

and sort the groups in descending order. Let $\{g_{t,1}, \ldots, g_{t,K}\}$ denote the indices of the top-K groups, where $K \ll G$.

B. Physics-Aware Wavenumber Tokenization

Instead of feeding raw complex CSI into the predictor, we construct a compact token sequence that preserves the dominant modal structure. For the kth selected group at time t, we define the token

$$\mathbf{z}_{t,k} = \mathbf{E}_{\mathrm{id}}(g_{t,k}) + \phi_e\big(\log(\widetilde{E}_{g_{t,k},t} + \varepsilon)\big) + \mathbf{E}_{\mathrm{pos}}(u_{t,k}, v_{t,k}), \tag{17}$$

where $\mathbf{E}_{\mathrm{id}}(\cdot)$ is a learnable group-index embedding, $\phi_e(\cdot)$ is an MLP that maps the scalar energy descriptor to the model dimension, and $\mathbf{E}_{\mathrm{pos}}(u_{t,k}, v_{t,k})$ denotes the embedding of the group location on the receive–transmit block grid. Here, $(u_{t,k}, v_{t,k})$ identify the row and column positions of group $g_{t,k}$ in the predefined group partition.

To retain global operating conditions, we further define a frame-level token

$$\mathbf{z}_t^{\mathrm{glo}} = \phi_g\Big(\big[\mathrm{SNR}_t,\, P,\, N_{\mathrm{RF}}^{(r)},\, N_{\mathrm{RF}}^{(t)},\, \|\widetilde{\mathbf{H}}_{a,t}\|_F^2\big]\Big)\,, \tag{18}$$

and, when adaptive probing is used, an additional probing token $\mathbf{z}_t^{\mathrm{prb}}$ that encodes the probing pattern identifier. The token matrix of time frame t is then formed as

$$\mathbf{Z}_t = \left[\mathbf{z}_t^{\mathrm{glo}}, \mathbf{z}_t^{\mathrm{prb}}, \mathbf{z}_{t,1}, \ldots, \mathbf{z}_{t,K}\right] \in \mathbb{R}^{N_{\mathrm{tok}} \times d}, \tag{19}$$

where d is the embedding dimension and $N_{\mathrm{tok}} = K + 2$ when the probing token is included.

C. Causal Transformer Predictor

Given the token sequence of the past L frames, $\{\mathbf{Z}_{t-L+1}, \ldots, \mathbf{Z}_t\}$, we employ a causal transformer to learn the temporal evolution of the groupwise wavenumber structure. Let

$$\mathbf{U}_t = \mathrm{Transformer}_\Theta(\mathbf{Z}_{t-L+1}, \ldots, \mathbf{Z}_t)\,, \tag{20}$$

where Θ denotes the trainable parameters. The final hidden state associated with the most recent frame is passed to two prediction heads:

$$\widehat{\mathbf{b}}_{t+1} = \sigma(\phi_b(\mathbf{U}_t)) \in [0,1]^G, \tag{21}$$

$$\widehat{\mathbf{e}}_{t+1} = \phi_e^{\mathrm{out}}(\mathbf{U}_t) \in \mathbb{R}^G, \tag{22}$$

where $\sigma(\cdot)$ is the element-wise sigmoid function. The first head produces the probability that each group is active at the next time instant, whereas the second head predicts the corresponding log-energy.

The proposed predictor is deliberately lightweight: it models only the low-dimensional support and energy trajectories, while the instantaneous high-resolution channel reconstruction remains delegated to the subsequent model-based recovery stage. This design substantially reduces the learning burden and improves robustness against noisy or imperfect historical estimates.

D. Training Objective

Let $\mathbf{b}_{t+1}$ and $\mathbf{e}_{t+1}$ denote the ground-truth support and log-energy labels obtained from (11) and (12). The predictor is trained by minimizing the composite loss

$$\mathcal{L} = \lambda_b \mathcal{L}_{\mathrm{BCE}} + \lambda_e \mathcal{L}_{\mathrm{Huber}}, \tag{23}$$

where

$$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{G}\sum_{g=1}^{G}\Big(b_{g,t+1}\log \widehat{b}_{g,t+1} + (1-b_{g,t+1})\log(1-\widehat{b}_{g,t+1})\Big), \tag{24}$$

$$\mathcal{L}_{\mathrm{Huber}} = \frac{1}{G}\sum_{g=1}^{G}\ell_{\mathrm{Huber}}(\widehat{e}_{g,t+1} - e_{g,t+1}). \tag{25}$$

The binary cross-entropy term encourages accurate support detection, whereas the Huber term stabilizes the regression of group energies under occasional large deviations. During deployment, the predictor takes as input only noisy historical measurements and coarse estimates, thereby matching the practical operating condition.

IV. Predictive Warm-Started Group-SBL Recovery

We now integrate the predicted support and energy states into a structured Bayesian recovery framework. The key idea is to use the LLM predictor to initialize the group activity hyperparameters, while the final channel estimate is still obtained by solving the physics-consistent inverse problem associated with (7).

A. Group-Structured Bayesian Prior

Let $\mathbf{h}_{t+1} = \mathrm{vec}(\mathbf{H}_{a,t+1})$ and partition it into G groups according to (9). We impose the Gaussian group-ARD prior

$$p(\mathbf{h}_{t+1} \mid \boldsymbol{\gamma}_{t+1}) = \mathcal{CN}(\mathbf{0}, \boldsymbol{\Gamma}_{t+1}), \tag{26}$$

where

$$\boldsymbol{\Gamma}_{t+1} = \mathrm{blkdiag}(\gamma_{1,t+1}\mathbf{I}_{d_1}, \ldots, \gamma_{G,t+1}\mathbf{I}_{d_G}), \tag{27}$$

$d_g = |\mathcal{I}_g|$, and $\gamma_{g,t+1} \geq 0$ controls the activity of the gth group. Small values of $\gamma_{g,t+1}$ suppress the entire group, whereas large values preserve it.

Conditioned on the measurement model (7), the likelihood is

$$p(\mathbf{y}_{t+1} \mid \mathbf{h}_{t+1}) = \mathcal{CN}\big(\mathbf{C}_{t+1}\mathbf{h}_{t+1}, \sigma_{t+1}^2\mathbf{I}\big). \tag{28}$$

Hence, the posterior remains Gaussian:

$$p(\mathbf{h}_{t+1} \mid \mathbf{y}_{t+1}, \boldsymbol{\gamma}_{t+1}, \sigma_{t+1}^2) = \mathcal{CN}(\boldsymbol{\mu}_{t+1}, \boldsymbol{\Sigma}_{t+1}), \tag{29}$$

with

$$\boldsymbol{\mu}_{t+1} = \boldsymbol{\Gamma}_{t+1}\mathbf{C}_{t+1}^H\boldsymbol{\Sigma}_{y,t+1}^{-1}\mathbf{y}_{t+1}, \tag{30}$$

$$\boldsymbol{\Sigma}_{t+1} = \boldsymbol{\Gamma}_{t+1} - \boldsymbol{\Gamma}_{t+1}\mathbf{C}_{t+1}^H\boldsymbol{\Sigma}_{y,t+1}^{-1}\mathbf{C}_{t+1}\boldsymbol{\Gamma}_{t+1}, \tag{31}$$

$$\boldsymbol{\Sigma}_{y,t+1} = \sigma_{t+1}^2\mathbf{I} + \mathbf{C}_{t+1}\boldsymbol{\Gamma}_{t+1}\mathbf{C}_{t+1}^H. \tag{32}$$

B. Predictive Hyperparameter Initialization

In conventional SBL, the hyperparameters are often initialized uniformly, which ignores the temporal structure of the channel. Here, we instead use the predicted group energies to initialize the prior:

$$\gamma_{g,t+1}^{(0)} = \mathrm{clip}(\exp(\widehat{e}_{g,t+1}), \gamma_{\min}, \gamma_{\max}), \qquad g = 1, \ldots, G, \tag{33}$$

where $0 < \gamma_{\min} < \gamma_{\max}$ are design parameters. To further incorporate the predicted support, we optionally apply

$$\gamma_{g,t+1}^{(0)} \leftarrow \gamma_{\min}, \qquad \text{if } \widehat{b}_{g,t+1} < \rho, \tag{34}$$

where $\rho \in (0,1)$ is a screening threshold. In this way, the predictor does not directly output the final channel estimate; rather, it provides a data-driven warm start for the Bayesian inference engine.

Given the initialization in (33)–(34), the group-SBL updates are iterated according to

$$\gamma_{g,t+1}^{\mathrm{new}} = \frac{1}{d_g}\left(\|\boldsymbol{\mu}_{g,t+1}\|_2^2 + \mathrm{tr}\big([\boldsymbol{\Sigma}_{t+1}]_{\mathcal{I}_g,\mathcal{I}_g}\big)\right), \tag{35}$$

where $\boldsymbol{\mu}_{g,t+1} = \boldsymbol{\mu}_{t+1}|_{\mathcal{I}_g}$. If desired, the noise variance can also be updated as

$$\sigma_{t+1}^{2\,\mathrm{new}} = \frac{1}{M}\left(\|\mathbf{y}_{t+1} - \mathbf{C}_{t+1}\boldsymbol{\mu}_{t+1}\|_2^2 + \mathrm{tr}\big(\mathbf{C}_{t+1}\boldsymbol{\Sigma}_{t+1}\mathbf{C}_{t+1}^H\big)\right). \tag{36}$$

C. Operator-Form Large-Scale Implementation

For realistic HMIMO dimensions, explicitly forming $\mathbf{C}_{t+1}$, $\boldsymbol{\Gamma}_{t+1}$, or $\boldsymbol{\Sigma}_{t+1}$ is computationally undesirable. We therefore retain the operator-form implementation throughout. Define the groupwise scaling operator $\mathcal{G}_{\boldsymbol{\gamma}}(\cdot)$ on a matrix $\mathbf{Z} \in \mathbb{C}^{L_R \times L_S}$ by

$$[\mathcal{G}_{\boldsymbol{\gamma}}(\mathbf{Z})]_{\ell,m} = \gamma_g[\mathbf{Z}]_{\ell,m}, \qquad (\ell,m) \in \mathcal{G}_g, \tag{37}$$

where $\mathcal{G}_g$ denotes the set of matrix indices belonging to group g. Then, for any compatible vector $\mathbf{r}$,

$$\mathbf{C}_{t+1}\boldsymbol{\Gamma}_{t+1}\mathbf{C}_{t+1}^H\mathbf{r} = P_{\mathrm{dim}}\,\mathcal{A}_{t+1}\big(\mathcal{G}_{\boldsymbol{\gamma}}\big(\mathcal{A}_{t+1}^*(\mathbf{r})\big)\big). \tag{38}$$

Hence, the linear system

$$\mathbf{u}_{t+1} = \boldsymbol{\Sigma}_{y,t+1}^{-1}\mathbf{y}_{t+1} \tag{39}$$

can be solved efficiently by conjugate-gradient iterations using only repeated applications of $\mathcal{A}_{t+1}(\cdot)$, $\mathcal{A}_{t+1}^*(\cdot)$, and $\mathcal{G}_{\boldsymbol{\gamma}}(\cdot)$.

Once $\mathbf{u}_{t+1}$ is obtained, the posterior mean estimate of the wavenumber-domain channel can be written in matrix form as

$$\widehat{\mathbf{H}}_{a,t+1} = \sqrt{P_{\mathrm{dim}}}\,\mathcal{G}_{\boldsymbol{\gamma}_{t+1}}\big(\mathcal{A}_{t+1}^*(\mathbf{u}_{t+1})\big). \tag{40}$$

The corresponding spatial-domain channel estimate is then reconstructed as

$$\widehat{\mathbf{H}}_{t+1} = \boldsymbol{\Psi}_R\widehat{\mathbf{H}}_{a,t+1}\boldsymbol{\Psi}_S^H. \tag{41}$$

D. Optional Safe Support Screening

To further reduce complexity, the predicted support can be used to form a screened active-set candidate:

$$\mathcal{S}_{t+1}^{\mathrm{pred}} = \big\{g : \widehat{b}_{g,t+1} \geq \rho\big\} \cup \mathcal{N}_\nu\Big(\big\{g : \widehat{b}_{g,t+1} \geq \rho\big\}\Big), \tag{42}$$

where $\mathcal{N}_\nu(\cdot)$ denotes a ν-hop neighborhood expansion on the group grid. The neighborhood expansion is included to mitigate the impact of occasional prediction errors near the support boundary. In implementation, groups outside $\mathcal{S}_{t+1}^{\mathrm{pred}}$ can either be fixed at $\gamma_{\min}$ or updated less frequently. This yields a safe screening mechanism that preserves robustness while lowering the computational load.

Overall, the proposed recovery pipeline consists of two coupled stages: first, a causal sequence model predicts the low-dimensional temporal state of the channel in the wavenumber domain; second, the predicted state is injected into a physics-consistent operator-form group-SBL solver to produce the final high-resolution channel estimate. This hybrid design retains interpretability, leverages temporal regularity, and remains scalable for large HMIMO systems.

REFERENCES

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